

Q:
$$a_{10}x^2 + 2a_{12}xy + a_{22}y^2 + a_{10}x + a_{20}y + a_{00} = 0$$

Q:
$$(x \ y) \cdot \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + (a_{10} \ a_{20}) \cdot \begin{pmatrix} x \\ y \end{pmatrix} + a_{00} = 0$$

If we build an orthonormal basis β of eigenvectors, then if $B = (v_1, v_2)$

$$v_{1,B} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \underbrace{v_{1,B}}_{\text{coordinates in initial basis}} \cdot \underbrace{\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}_{\text{coordinates in the basis of eigenvectors}}$$

Q:
$$(x \ y) \cdot \underbrace{\begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix}}_{\substack{= \frac{1}{11}(B) \cdot \frac{1}{11}(B) \\ = \frac{1}{11} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{11}}} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + (a_{10} \ a_{20}) \cdot \underbrace{v_{1,B}}_{\substack{= \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ = \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + a_{00} = 0$$

Q:
$$x_1^2 x_2^2 + 2x_1^2 + \beta x_1 + a_{00} = 0$$

We choose $v_{1,B}$ so that $v_{1,B} \in SO(2)$

The next step is a translation

We force squares: $3x_1^2 + 2 \cdot \frac{1}{2\lambda_1} x_1^2 + \frac{1}{4\lambda_1^2} + 2 \left(\frac{1}{4\lambda_1^2} + \frac{\beta}{2\lambda_1} x_1 + \frac{\beta^2}{4\lambda_1^2} \right) + a_{00} - \frac{1}{4\lambda_1^2} - \frac{\beta^2}{4\lambda_1^2} = 0$

$$\Leftrightarrow \lambda_1 \left(x_1 + \frac{\beta}{2\lambda_1} \right)^2 + \lambda_1 \left(\frac{\beta}{2\lambda_1} + \frac{\beta}{2\lambda_1} \right)^2 + k = 0$$

$$\begin{cases} x_1 = x + \frac{\beta}{2\lambda_1} \\ y_1 = y + \frac{\beta}{2\lambda_1} \end{cases}$$

$$\Rightarrow Q: \lambda_1 x_1^2 + \lambda_2 y_1^2 = -h$$

Q:
$$\frac{x_1^2}{\frac{-h}{\lambda_1}} + \frac{y_1^2}{\frac{-h}{\lambda_2}} = 1$$

$$\begin{cases} \frac{x_1^2}{5} + \frac{y_1^2}{\sqrt{11}} \\ y_1 = \frac{y_2}{\sqrt{\frac{1}{11}}} \end{cases}$$

4. For each of the following equations write down the associated matrix and bring the equation in canonical form.

a) $-x^2 + xy - y^2 = 0$.

b) $6xy + x - y = 0$.

b) $6xy + x - y = 0$

$$A(2) = \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$$

$$\lambda_1 = 3, \lambda_2 = -3$$

$$\begin{pmatrix} -3 & 3 \\ 3 & -3 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} -3x + 3y = 0 \\ 3x - 3y = 0 \end{cases} \Leftrightarrow y = x \Rightarrow \text{So } V(\lambda_1) = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} \mid x \in \mathbb{R} \right\} = \langle (1, 1) \rangle$$

We chose $v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$$V(\lambda_2) = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} \mid x \in \mathbb{R} \right\} = \langle (1, -1) \rangle$$

We chose $v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$B = (v_1, v_2) = \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) \text{ orthonormal basis of } \mathbb{R}^2$$

$$v_{1,B} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

But $\det v_{1,B} = -1$. We want the transformation to be rotation

We chose $v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$$v_{1,B} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \in SO(2)$$

Q:
$$6xy + x - y = 0$$

$$(x \ y) \cdot \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix} + (1 \ -1) \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

Q:
$$(x \ y) \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix} \cdot \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + (1 \ -1) \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$A = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix} \cdot \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & -3 \end{pmatrix}$$

$$(1 \ -1) \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = (0 \ -2)$$

Q:
$$(x \ y) \cdot \begin{pmatrix} 3 & 0 \\ 0 & -3 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -2 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = c \Rightarrow$$

$$\Rightarrow Q: 3x_1^2 - 3y_1^2 - \sqrt{2}y_1 = 0$$

Q:
$$3x_1^2 - 3y_1^2 + 2 \cdot \frac{\sqrt{2}}{2} y_1 + \frac{1}{18} = -\frac{1}{6}$$

$$3x_1^2 - 3 \left(y_1 + \frac{\sqrt{2}}{6} \right)^2 = -\frac{1}{6}$$

$$\begin{cases} x_1 = x_2 \\ y_1 = y_2 + \frac{\sqrt{2}}{6} \end{cases}$$

Q:
$$3x_1^2 - 3y_1^2 = -\frac{1}{6}$$

$$-\frac{x_1^2}{\frac{1}{18}} + \frac{y_1^2}{\frac{1}{18}} = 1 \Rightarrow \text{hyperbola that is oriented}$$

$$\begin{cases} x_1 = x_2 \\ y_1 = y_2 \end{cases}$$

$$\frac{x_2^2}{-\frac{1}{18}} + \frac{y_2^2}{\frac{1}{18}} = 1$$

Not necessarily $\frac{x_2}{\sqrt{18}}$

$$x_2 = \frac{x_3}{\sqrt{18}}, y_2 = \frac{y_3}{\sqrt{18}}$$

$$\Rightarrow x_3^2 - y_3^2 = 1$$

5. In each of the following cases, decide the type of the quadratic curve based on the parameter $a \in \mathbb{R}$.

a) $x^2 - 4xy + y^2 = a$.

b) $x^2 + 4xy + y^2 = a$.

a) $x^2 - 4xy + y^2 - 2x + 1 = a$

$$(x^2 - 4xy + 4y^2) - 3y^2 - 4y - 2x + 1 = a$$

$$(x - 2y)^2 - 3y^2 + 4y - 2x + 1 = a$$

$$x_1 = x - 2y, y_1 = x - 2y + y$$

Q:
$$x_1^2 - 3y_1^2 - 2x_1 + 1 = a$$

$$x_1^2 - 2x_1 + 1 - 3y_1^2 = a$$

$$(x_1 - 1)^2 - 3y_1^2 = a$$

$$\begin{cases} x_1 = x_2 - 1 \\ y_1 = y_2 \end{cases}$$

$$x_2^2 - 3y_2^2 = a \begin{cases} a > 0: \text{hyperbola (x-aligned)} \\ a = 0: \text{ellipses, } x_2 = \pm \sqrt{3} y_2 \\ a < 0: \text{hyperbola (y-aligned)} \end{cases}$$

Lagrange method

7. Find the canonical equation for each of the following cases

a) $5x^2 + 4xy + 8y^2 - 32x - 56y + 80 = 0$,

b) $8y^2 + 6xy - 12x - 26y + 11 = 0$,

c) $x^2 - 4xy + y^2 - 6x + 2y + 1 = 0$.

a) $5x^2 + 2xy + 2y^2 + 8y^2 - 32x - 56y + 80 = 0$

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 8 \end{pmatrix}$$

$$a_{10} = -32$$

$$a_{20} = -56$$

$$\begin{vmatrix} 5-\lambda & 2 \\ 2 & 8-\lambda \end{vmatrix} = 0 \Leftrightarrow (5-\lambda)(8-\lambda) - 4 = 0$$

$$40 - 8\lambda - 8\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 16\lambda + 36 = 0$$

$$\Delta = 16^2 - 4 \cdot 36 = 25$$

$$\lambda_1 = \frac{16 + 5}{2} = 9$$

$$\lambda_2 = \frac{16 - 5}{2} = 4$$

$$\lambda_1 = 9: A - \lambda_1 I = \begin{pmatrix} -4 & 2 \\ 2 & -1 \end{pmatrix} \Rightarrow \begin{cases} -4x + 2y = 0 \\ 2x - y = 0 \end{cases} \Rightarrow V(\lambda_1) = \langle (1, 2) \rangle$$

We chose $v_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$\lambda_2 = 4: A - \lambda_2 I = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \Rightarrow \begin{cases} x + 2y = 0 \\ 2x + 4y = 0 \end{cases} \Rightarrow V(\lambda_2) = \langle (-2, 1) \rangle$$

We chose $v_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$$v_{1,B} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$$

Q:
$$(x \ y) \cdot \underbrace{\begin{pmatrix} 5 & 2 \\ 2 & 8 \end{pmatrix}}_{\substack{= \frac{1}{5} \cdot \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \\ = \frac{1}{5} \cdot \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \cdot \frac{1}{5}}} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + (a_{10} \ a_{20}) \cdot \underbrace{v_{1,B}}_{\substack{= \frac{1}{\sqrt{5}} \cdot \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \\ = \frac{1}{\sqrt{5}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + a_{00} = 0$$

$$(x_1 \ y_1) \cdot \begin{pmatrix} 9 & 0 \\ 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} - \frac{8}{\sqrt{5}} \begin{pmatrix} 4 & 7 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + 80 = 0$$

$$9x_1^2 + 4y_1^2 - \frac{8}{\sqrt{5}} \begin{pmatrix} 4 & 7 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + 80 = 0$$

$$9x_1^2 + 4y_1^2 - \frac{144\sqrt{5}}{5} x_1 + \frac{8\sqrt{5}}{5} y_1 + 80 = 0$$

$$(3x_1 - 2\sqrt{5})^2 + 3y_1^2 + \frac{2y_1^2}{5} + \left(\frac{2y_1}{5} \right)^2 + \frac{2\sqrt{5}}{5} \cdot 2y_1 + \frac{4}{5} + 20 = 0$$

$$\left(3x_1 - 2\sqrt{5} \right)^2 + \left(2y_1 - 2\sqrt{5} \right)^2 = 36$$

$$\begin{cases} x_2 = 3x_1 - 2\sqrt{5} \\ y_2 = 2y_1 - 2\sqrt{5} \end{cases}$$

$$x_2^2 + y_2^2 = 36 \quad | :36$$

$$\frac{x_2^2}{36} + \frac{y_2^2}{36} = 1$$

b) $8y^2 + 3xy + 24x + 0y^2 - 12x - 26y + 11 = 0$

$$A = \begin{pmatrix} 8 & 3 \\ 3 & 0 \end{pmatrix}$$

$$\begin{vmatrix} 8-\lambda & 3 \\ 3 & -\lambda \end{vmatrix} = 0 \Leftrightarrow \lambda(8-\lambda) - 9 = 0$$

$$(\lambda - 4)(\lambda - 9) - 9 = 0$$

$$(2 - 4)^2 - 16 - 9 = 0$$

$$(2 - 4) = 5 \Rightarrow \lambda_1 = -1 \Rightarrow \text{We have a hyperbola}$$

$$\lambda_2 = 9$$

$$\lambda_1 = -1: A - \lambda_1 I = \begin{pmatrix} 9 & 3 \\ 3 & 1 \end{pmatrix} \Rightarrow \begin{cases} 9x + 3y = 0 \\ 3x + y = 0 \end{cases} \Rightarrow V(\lambda_1) = \langle (-1, 3) \rangle$$

We chose $v_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

$$\lambda_2 = 9: A - \lambda_2 I = \begin{pmatrix} -1 & 3 \\ 3 & -9 \end{pmatrix} \Rightarrow \begin{cases} -x + 3y = 0 \\ 3x - 9y = 0 \end{cases} \Rightarrow V(\lambda_2) = \langle (3, 1) \rangle$$

We chose $v_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} -3 \\ 1 \end{pmatrix}$

$$v_{1,B} = \frac{1}{\sqrt{10}} \begin{pmatrix} -1 & -3 \\ 3 & 1 \end{pmatrix}$$

Q:
$$(x \ y) \cdot \underbrace{\begin{pmatrix} 8 & 3 \\ 3 & 0 \end{pmatrix}}_{\substack{= \frac{1}{10} \cdot \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \\ = \frac{1}{10} \cdot \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \cdot \frac{1}{10}}} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + (a_{10} \ a_{20}) \cdot \underbrace{v_{1,B}}_{\substack{= \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} -1 & -3 \\ 3 & 1 \end{pmatrix} \\ = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + a_{00} = 0$$

b) $\begin{vmatrix} 5 & -13 \\ -15 & 5 \end{vmatrix} = 25 + 195 = 200$

$$\begin{vmatrix} 5-\lambda & -13 \\ -15 & 5-\lambda \end{vmatrix} = 0 \Leftrightarrow (5-\lambda)^2 - 195 = 0$$

$$\lambda^2 - 10\lambda + 25 - 195 = 0$$

$$\lambda^2 - 10\lambda - 170 = 0$$

$$\Delta = 100 + 4 \cdot 170 = 780$$

$$\lambda_1 = \frac{10 + \sqrt{780}}{2} = \frac{10 + 2\sqrt{195}}{2} = 5 + \sqrt{195}$$

$$\lambda_2 = 5 - \sqrt{195}$$

$$\lambda_1 = 5 + \sqrt{195}$$

$$\begin{vmatrix} \sqrt{195} & -13 \\ -15 & \sqrt{195} \end{vmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} \sqrt{195}x - 13y = 0 \\ -15x + \sqrt{195}y = 0 \end{cases} \Leftrightarrow \begin{cases} \sqrt{195}x - 13y = 0 \\ x = \frac{\sqrt{195}}{15}y \end{cases} \Leftrightarrow \begin{cases} 0 = 0 \\ x = \frac{\sqrt{195}}{15}y \end{cases} \Rightarrow V(\lambda_1) = \langle \left(\frac{\sqrt{195}}{15}, 1 \right) \rangle$$

We chose $v_1 = \frac{\sqrt{195}}{5} \begin{pmatrix} \frac{\sqrt{195}}{15} \\ 1 \end{pmatrix}$

Q:
$$a_{11}x^2 + 2a_{12}xy + a_{22}y^2 + a_{10}x + a_{20}y + a_{00} = 0$$

Steps: $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$

Step 2: Find eigenvalues

Step 3: Find eigenvectors (normalized)

$$v_{1,B} = (v_1 \ v_2)$$

Step 4: Q:
$$(x \ y) \cdot \underbrace{\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}}_{\substack{= \frac{1}{10} \cdot \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \\ = \frac{1}{10} \cdot \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \cdot \frac{1}{10}}} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + (a_{10} \ a_{20}) \cdot \underbrace{v_{1,B}}_{\substack{= \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} -1 & -3 \\ 3 & 1 \end{pmatrix} \\ = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + a_{00} = 0$$

$$(x_1 \ y_1) \cdot \begin{pmatrix} 9 & 0 \\ 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + (a_{10} \ a_{20}) \cdot \underbrace{v_{1,B}}_{\substack{= \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} -1 & -3 \\ 3 & 1 \end{pmatrix} \\ = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}} \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + a_{00} = 0$$

$$9x_1^2 + 4y_1^2 + 2x_1 + \beta y_1 + a_{00} = 0$$

Steps: Force squares

$$9(x_1^2 + 2 \cdot \frac{1}{2\lambda_1} x_1^2 + \frac{1}{4\lambda_1^2}) + 2(x_1^2 + 2 \cdot \frac{\beta}{2\lambda_2} y_1 + \frac{\beta^2}{4\lambda_2^2}) + h$$

$$h = a_{00} - \frac{1}{4\lambda_1^2} - \frac{\beta^2}{4\lambda_2^2}$$

Step 5: $x_2 = x_1 + \frac{1}{2\lambda_1}$

$$\begin{cases} x_2 = x_1 + \frac{1}{2\lambda_1} \\ y_2 = y_1 + \frac{1}{2\lambda_2} \end{cases}$$

$$9x_2^2 + 4y_2^2 = -h$$

$$\frac{9x_2^2}{9} + \frac{4y_2^2}{4} = -h \quad | \cdot \frac{1}{-h}$$

$$-\frac{x_2^2}{h} + \frac{y_2^2}{h} = 1$$

6. Consider the rotation R_{θ} of $\mathbb{R}^2$ around the origin and the translation T_a of $\mathbb{R}^2$ with vector $v(1,0)$.

a) Give the algebraic form of the isometries R_{θ} , T_a and $T_a \circ R_{\theta}$.

b) Determine the equations of the hyperbola $\mathcal{H}: \frac{x^2}{9} - \frac{y^2}{9} - 1 = 0$ and the parabola $\mathcal{P}: y^2 - 8x = 0$ after transforming them with R_{θ} and with $T_a \circ R_{\theta}$ respectively.

$$R_{30}(\theta) = \begin{pmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

$$T_a(v) = \begin{pmatrix} x_p \\ y_p \end{pmatrix} + \begin{pmatrix$$